

## SAFETY DATA SHEET

Creation Date 20-October-2009

Revision Date 29-March-2024

Revision Number 5

### 1. Identification

**Product Name** Iodine

**Cat No. :** A12278

**CAS-No** 7553-56-2

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                                        | Category 4 |
| <b>Acute dermal toxicity</b>                                      | Category 4 |
| <b>Acute Inhalation Toxicity</b>                                  | Category 4 |
| <b>Skin Corrosion/Irritation</b>                                  | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                          | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b>           | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |
| <b>Specific target organ toxicity - (repeated exposure)</b>       | Category 1 |
| Target Organs - Kidney, Liver, Blood, Thyroid.                    |            |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled  
Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation  
Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
Take off contaminated clothing and wash it before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic organisms

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Iodine    | 7553-56-2 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if

symptoms occur.

**Most important symptoms/effects** None reasonably foreseeable.  
**Notes to Physician** Treat symptomatically

## 5. Fire-fighting measures

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available  
**Explosion Limits**

**Upper** No data available  
**Lower** No data available  
**Sensitivity to Mechanical Impact** No information available  
**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Hydrogen iodide.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store in metal containers. Keep at temperatures below 25°C. Incompatible Materials. Strong oxidizing agents. Finely powdered metals. Ammonia. Alcohols. copper.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                                          | British Columbia | Ontario TWAEV                  | Quebec                           | ACGIH TLV                      | OSHA PEL                                         | NIOSH                           |
|-----------|--------------------------------------------------|------------------|--------------------------------|----------------------------------|--------------------------------|--------------------------------------------------|---------------------------------|
| Iodine    | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> | Ceiling: 0.1 ppm | TWA: 0.01 ppm<br>STEL: 0.1 ppm | Ceiling: 0.1 ppm<br>Ceiling: 1.0 | TWA: 0.01 ppm<br>STEL: 0.1 ppm | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup> | IDLH: 2 ppm<br>Ceiling: 0.1 ppm |

|  |  |  |  |                   |  |                                                                            |                              |
|--|--|--|--|-------------------|--|----------------------------------------------------------------------------|------------------------------|
|  |  |  |  | mg/m <sup>3</sup> |  | (Vacated)<br>Ceiling: 0.1 ppm<br>(Vacated)<br>Ceiling: 1 mg/m <sup>3</sup> | Ceiling: 1 mg/m <sup>3</sup> |
|--|--|--|--|-------------------|--|----------------------------------------------------------------------------|------------------------------|

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                     |                            |
|---------------------|----------------------------|
| Physical State      | Solid                      |
| Appearance          | Grey                       |
| Odor                | pungent                    |
| Odor Threshold      | No information available   |
| pH                  | 5.1 saturated solution     |
| Melting Point/Range | 113 °C / 235.4 °F          |
| Boiling Point/Range | 185 °C / 365 °F @ 760 mmHg |
| Flash Point         | No information available   |

|                                        |                          |
|----------------------------------------|--------------------------|
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 0.41 hPa @ 25 °C         |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Bulk Density                           | ~ 2100 kg/m <sup>3</sup> |
| Solubility                             | slightly soluble         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | I <sub>2</sub>           |
| Molecular Weight                       | 253.81                   |

## 10. Stability and reactivity

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                 |
| Stability                        | Stable under normal conditions.                                            |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Excess heat.                  |
| Incompatible Materials           | Strong oxidizing agents, Finely powdered metals, Ammonia, Alcohols, copper |
| Hazardous Decomposition Products | Hydrogen iodide                                                            |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                   |
| Hazardous Reactions              | None under normal processing.                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component | LD50 Oral         | LD50 Dermal           | LC50 Inhalation       |
|-----------|-------------------|-----------------------|-----------------------|
| Iodine    | 315 mg/kg ( Rat ) | 1425 mg/kg ( Rabbit ) | 4.588 mg/L 4h ( Rat ) |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes, respiratory system and skin                                          |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Iodine    | 7553-56-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)  
**STOT - repeated exposure** Kidney Liver Blood Thyroid

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae     | Freshwater Fish      | Microtox           | Water Flea           |
|-----------|----------------------|----------------------|--------------------|----------------------|
| Iodine    | EC50 = 0.13 mg/L 72h | LC50 = 1.67 mg/L 96h | EC50 = 280 mg/L 3h | EC50 = 0.55 mg/L 48h |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component | log Pow |
|-----------|---------|
| Iodine    | 2.49    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3495  
 Proper Shipping Name IODINE  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group III

### TDG

UN-No UN3495  
 Proper Shipping Name IODINE  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group III

### IATA

UN-No UN3495  
 Proper Shipping Name IODINE  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group III

### IMDG/IMO

UN-No UN3495  
 Proper Shipping Name IODINE  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group III

## 15. Regulatory information

## International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Iodine    | 7553-56-2 | X   | -    | X    | ACTIVE                                        | 231-442-4 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Iodine    | 7553-56-2 | X     | KE-21023 | X    | -    | X    | X    | X     | X     |

## Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Iodine    | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Iodine    | 7553-56-2 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Iodine    | 7553-56-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Product Safety Department  
Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Creation Date</b>    | 20-October-2009                                    |
| <b>Revision Date</b>    | 29-March-2024                                      |
| <b>Print Date</b>       | 29-March-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

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**End of SDS**